Project - High Level Design on Autonomous Energy Researcher Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

The **Autonomous Energy Researcher Agent** is an LLM-powered multi-agent system designed to automate research in the energy domain, including:

- Renewable energy trends
- Oil & gas market analysis
- Power sector developments
- Sustainability and climate policies
- Energy investment insights

The system uses:

- Large Language Models (LLMs)
- LangChain framework
- LangGraph for graph-based orchestration
- Web search tools for real-time data retrieval

The system autonomously performs reasoning, tool usage, and structured report generation.

1.1 Scope of the Document

This High-Level Design (HLD) document defines the overall architecture and structural design of the Autonomous Energy Researcher Agent system. It describes the system's graph-based orchestration using LangGraph, the integration of Large Language Models through LangChain, and the interaction with external web search tools for real-time energy data retrieval. The document outlines major components, system workflows, architecture diagrams, data handling mechanisms, integration points, and key design decisions that shape the solution. It also highlights non-functional requirements such as security, performance, scalability, and maintainability. This document focuses on high-level architectural structure and system behavior rather than low-level implementation details or source code.

1.2 Intended Audience

1 Project Evaluators

This document helps project evaluators understand the overall architecture, workflow, and design decisions of the Autonomous Energy Researcher Agent. Itprovides clarity on how the system meets academic and technical objectives defined in the course.

2 Technical Reviewers

Technical reviewers can use this document to analyze the architectural soundness, graph-based orchestration using LangGraph, and integration strategy with LangChain. It enables them to assess scalability, modularity, and system robustness.

3 Development Team Members

The development team can refer to this HLD to understand component responsibilities, system flow, and module interactions. It serves as a guiding blueprint for implementation, testing, and future enhancements.

4 Academic Assessment Panel

The academic panel can evaluate whether the project demonstrates proper system design principles, architectural clarity, and structured documentation. It also helps them assess innovation, technical depth, and alignment with the defined project scope.

1.3 System Overview

The system follows a **graph-based agent architecture** using LangGraph where:

- User submits an energy query
- A reasoning node decides next action
- Tool nodes fetch real-time data
- Analysis node structures findings
- Summary node generates final research report
- Output stored in repository

Unlike linear orchestration, LangGraph enables:

- Conditional branching
- Iterative reasoning
- Tool calling loops
- State-based execution

2. System Design

2.1 Application Design

The system follows a layered + graph hybrid architecture:

1 Presentation Layer

- Accepts user query
- Displays structured research output

2 Orchestration Layer – LangGraph

- Graph-based execution engine
- Node-based state transitions
- Conditional routing
- Tool integration

3 Agent Logic Layer

Nodes include:

- Planner Node
- Research Node
- Tool Node (Web Search)
- Analysis Node
- Summary Node

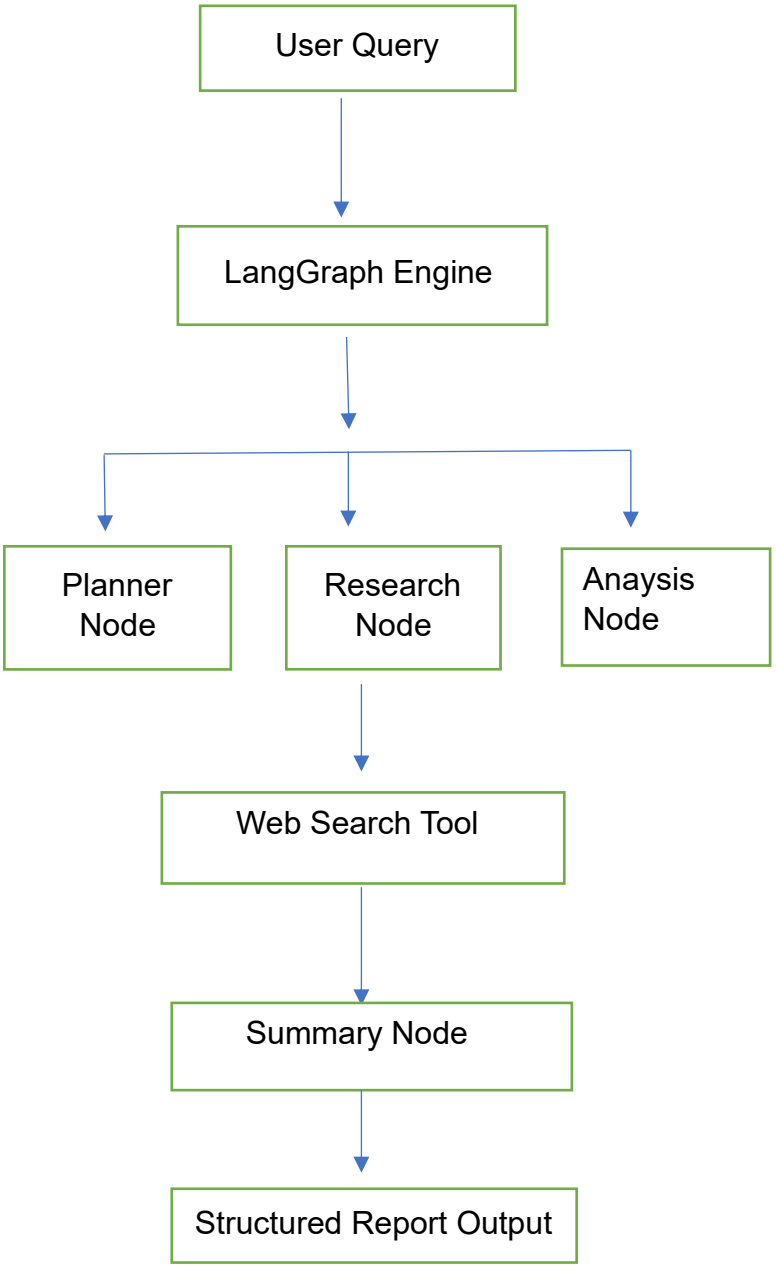
4 Tool Layer

- Web Search API
- LLM API

5 Storage Layer

- File-based structured research reports

2.2 Architecture Diagram (Conceptual)



The system is event-driven and state-controlled via LangGraph.

2.3 Process Workflow

1. User submits energy-related query
2. Planner Node interprets intent
3. Graph determines next node

4. Research Node calls Web Search Tool
5. Retrieved data stored in graph state
6. Analysis Node filters and validates data
7. Summary Node generates structured report
8. Final output returned and stored

This ensures dynamic reasoning instead of static sequencing.

2.4 Information Flow

User Query
→ LangGraph Entry Node
→ Planner
→ Tool Execution
→ State Update
→ Analysis
→ Summary
→ Output Storage

State object carries:

- Query
- Retrieved data
- Intermediate reasoning
- Final report

2.5 Components Design

1. Planner Node

- Intent detection
- Query decomposition
- Task planning

2. Research Node

- Real-time data retrieval
- Multiple source aggregation

3. Tool Node

- Web search execution
- External API calls

4. Analysis Node

- Deduplication
- Relevance filtering
- Insight extraction

5. Summary Node

- Structured formatting
- Executive summary creation
- Professional output

6. Storage Module

- File-based persistent storage
- Organized report directory

2.6 Key Design Decisions

1 Why LangGraph instead of CrewAI?

- Supports conditional execution
- Enables stateful workflows
- Better suited for tool-calling loops
- Allows dynamic reasoning paths

2 Graph-Based Orchestration

Chosen because energy research often requires:

- Iterative refinement
- Conditional branching
- Multi-step reasoning

3 Stateless External Interface

- Each query independent
- Avoids session complexity
- Simplifies scaling

4 File-Based Storage

- Lightweight
- Easy deployment
- Minimal infrastructure overhead

2.7 API Catalogue

1 LLM API

Purpose:

- Reasoning
- Planning
- Summarization
- Structured generation

2 Web Search API

Purpose:

- Real-time energy market data
- News & policy updates
- Industry insights

Integrated using LangChain tool wrappers.

3. Data Design

3.1 Data Model

Primary Entities:

- User Query
- Graph State Object

- Retrieved Web Data
- Processed Insights
- Final Report

Data Type: Unstructured text + structured state dictionary.

3.2 Data Access Mechanism

- Python-based file handling
- JSON/state-based transitions inside LangGraph
- Organized report storage

3.3 Data Retention Policies

- No long-term user tracking
- API logs limited
- Secure environment variable storage
- Optional log cleanup

3.4 Data Migration

Currently minimal requirement.
Future DB integration would include:

- Schema versioning
- Backup policies
- Staging validation

4. Interfaces

User Interface

- CLI or simple web interface
- Accepts textual input
- Displays structured research output

External Interfaces

- LLM API
- Web Search API
- Integrated via LangChain

5. State and Session Management

The system uses:

- LangGraph state object
- Controlled node transitions
- No persistent user session

Graph maintains internal state per execution cycle.

6. Caching

Current:

- No caching implemented
- Every query performs fresh reasoning

Future Enhancement:

- Cache repeated queries
- Token optimization
- Result memoization

7. Non-Functional Requirements

7.1 Security Aspects

- API keys stored in .env
- No hardcoded credentials
- Controlled file access
- Minimal exposure surface

7.2 Performance Aspects

- Target response time: 5–10 seconds
- Efficient node transitions
- Controlled token usage
- Optimized tool invocation
- Lightweight storage

8. References

- OpenAI – GPT Technical Report
- LangChain Documentation
- LangGraph Documentation
- ReAct Paper (Yao et al., 2023)
- AutoGPT Paper (Wu et al., 2023)